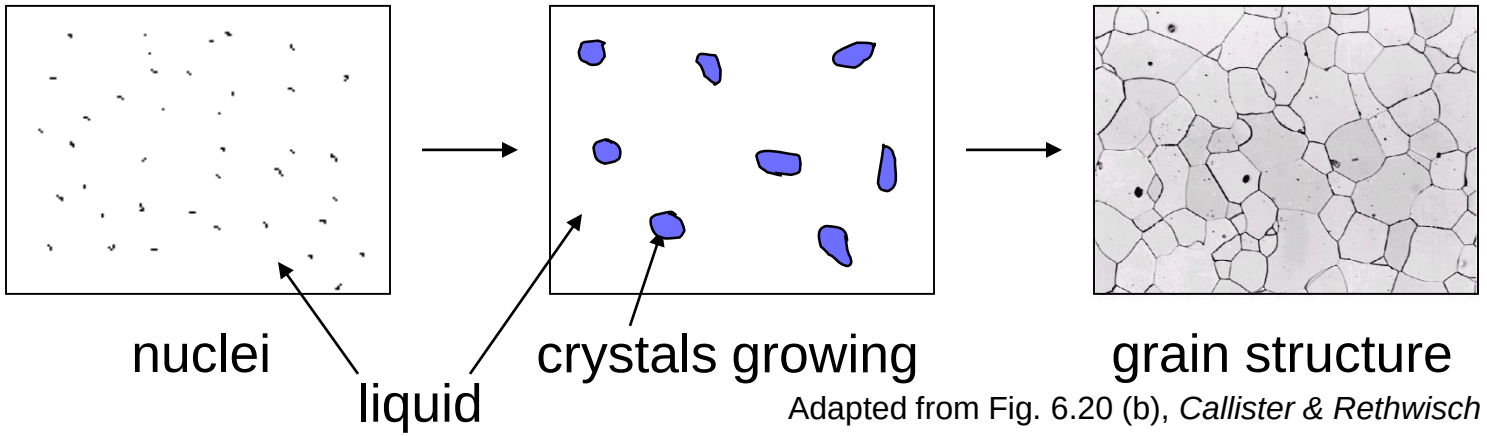
Chapter 6: Imperfections in Solids

ISSUES TO ADDRESS...

- What are the solidification mechanisms?
- What types of defects arise in solids?
- Can the number and type of defects be varied and controlled?
- How do defects affect material properties?
- Are defects undesirable?

Imperfections in Solids

- Solidification- result of casting of molten material
  - 2 steps
    - Nuclei form
    - Nuclei grow to form crystals – grain structure
- Start with a molten material – all liquid



[Photomicrograph courtesy of L. C. Smith and C. Brady, the National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD.)]

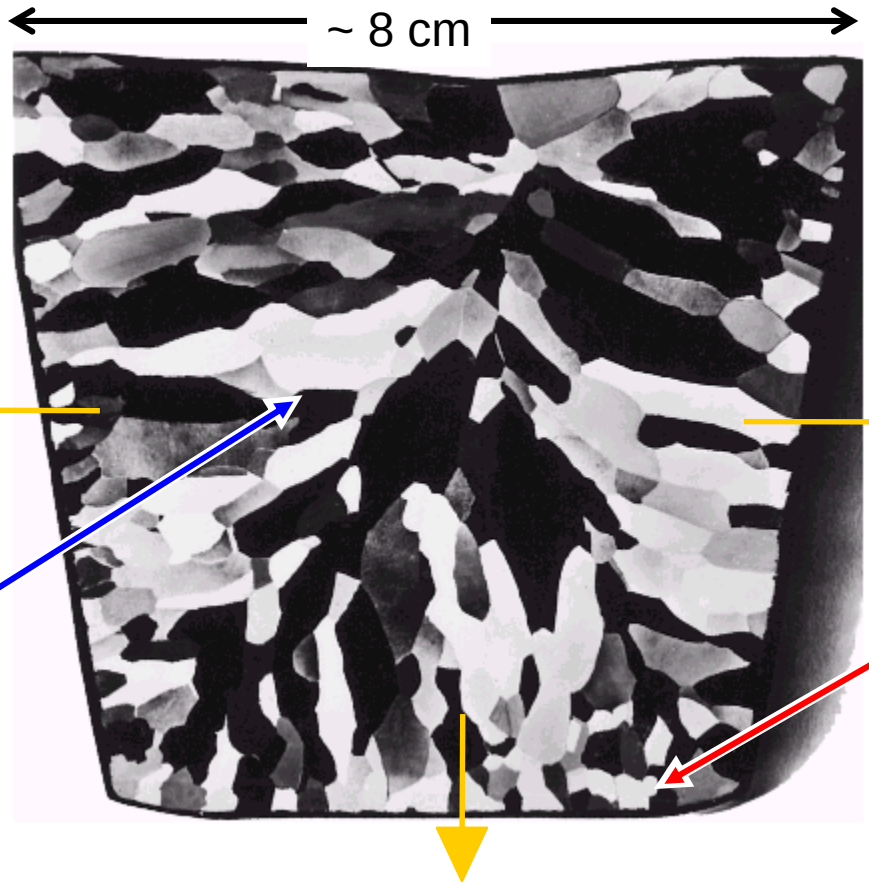
Adapted from Fig. 6.20 (b), *Callister & Rethwisch 9e.*

- Crystals grow until they meet each other

Solidification

Grains can be - equiaxed (roughly same size in all directions)

- columnar (elongated grains)



Adapted from Fig. 5.17, *Callister & Rethwisch 3e.* (Reproduced with permission from Metals Handbook, Vol. 9, 9th edition, Metallography and Microstructures, ASM International, Materials Park, OH, 1985.)

Grain Refiner - added to make smaller, more uniform, equiaxed grains.

Imperfections in Solids

There is no such thing as a perfect crystal.

- What are these imperfections?
- Why are they important?

Many of the important properties of materials are due to the presence of imperfections.

Types of Imperfections

- Vacancy atoms
- Interstitial atoms
- Substitutional atoms

Point defects

- Dislocations

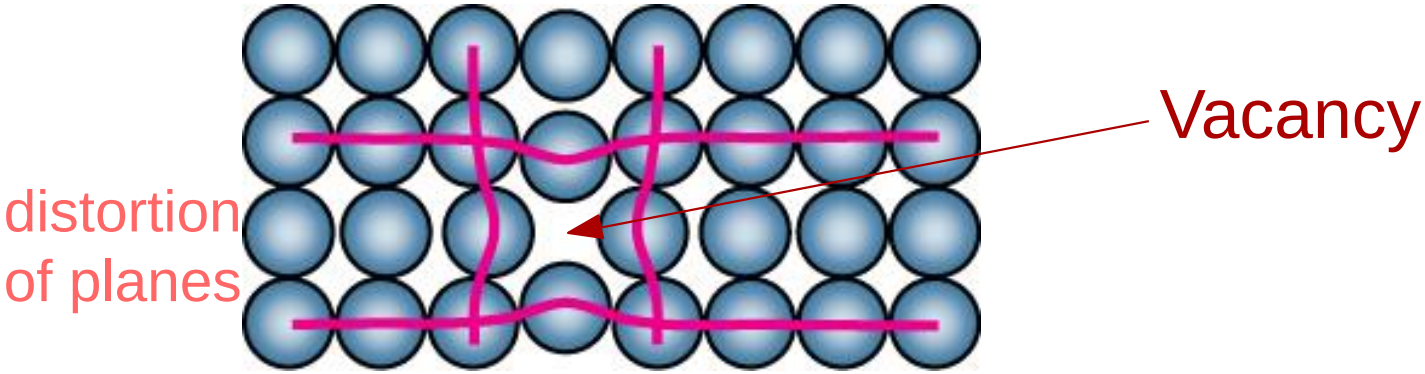
Line defects

- Grain Boundaries

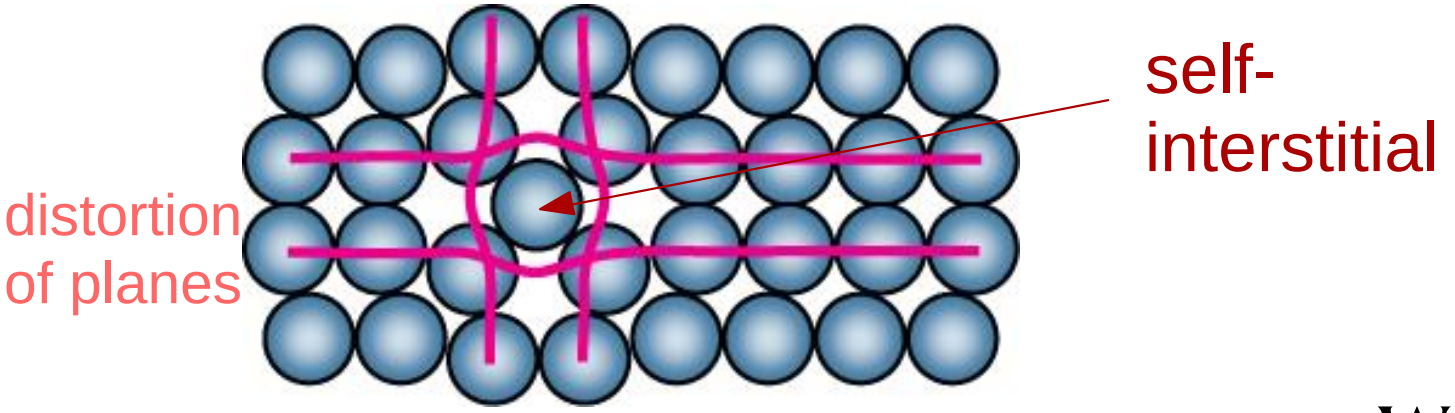
Area defects

Point Defects in Metals

- Vacancies:
  -vacant atomic sites in a structure.



- Self-Interstitials:
  -"extra" atoms positioned between atomic sites.



Equilibrium Concentration: Point Defects

- Equilibrium concentration varies with temperature!

No. of defects

Activation energy

$$\frac{N_v}{N} = \exp\left(\frac{-Q_v}{kT}\right)$$

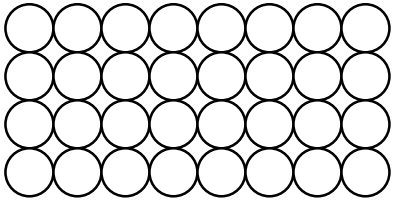
No. of potential defect sites

Temperature

Boltzmann's constant

(1.38 x 10^{-23} J/atom-K)

(8.62 x 10^{-5} eV/atom-K)



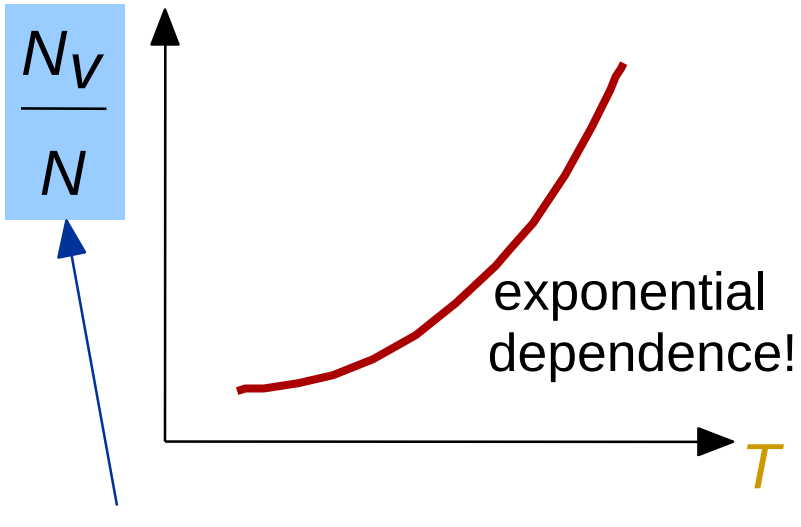
Each lattice site is a potential vacancy site

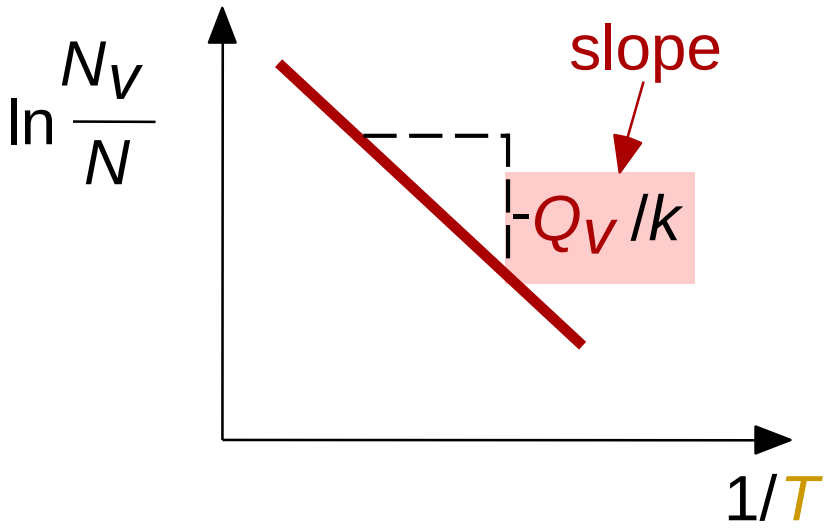
Measuring Activation Energy

- We can get Q_v from an experiment.
- Measure this...

$$\frac{N_V}{N} = \exp\left(\frac{-Q_v}{kT}\right)$$

- Replot it...





Estimating Vacancy Concentration

- Find the equil. No. of vacancies in 1 m^3 of Cu at 1000°C.
- Given:

$\rho = 8.4\ g/cm^3$ *Cu Mass No.* $= A_{Cu} = 63.5\ g/mol$

$Q_V = 0.9\ eV/atom$ $N_A = 6.02 \times 10^{23}$ atoms/mol

$$\frac{N_V}{N} = \exp\left(\frac{-Q_V}{kT}\right) = 2.7 \times 10^{-4}$$

0.9 eV/atom

1273 K

8.62×10^{-5} eV/atom-K

For 1 m^3, $N = \rho \times \frac{N_A}{A_{Cu}} \times 1\ m^3 = 8.0 \times 10^{28}$ sites

- Answer:

$N_V = (2.7 \times 10^{-4})(8.0 \times 10^{28})$ sites $= 2.2 \times 10^{25}$ vacancies

Point Defects

- Vacancies
  -- vacancies exist in ceramics for both cations and anions
- Interstitials
  -- interstitials exist for cations
  -- interstitials are not normally observed for anions because anions are large relative to the interstitial sites

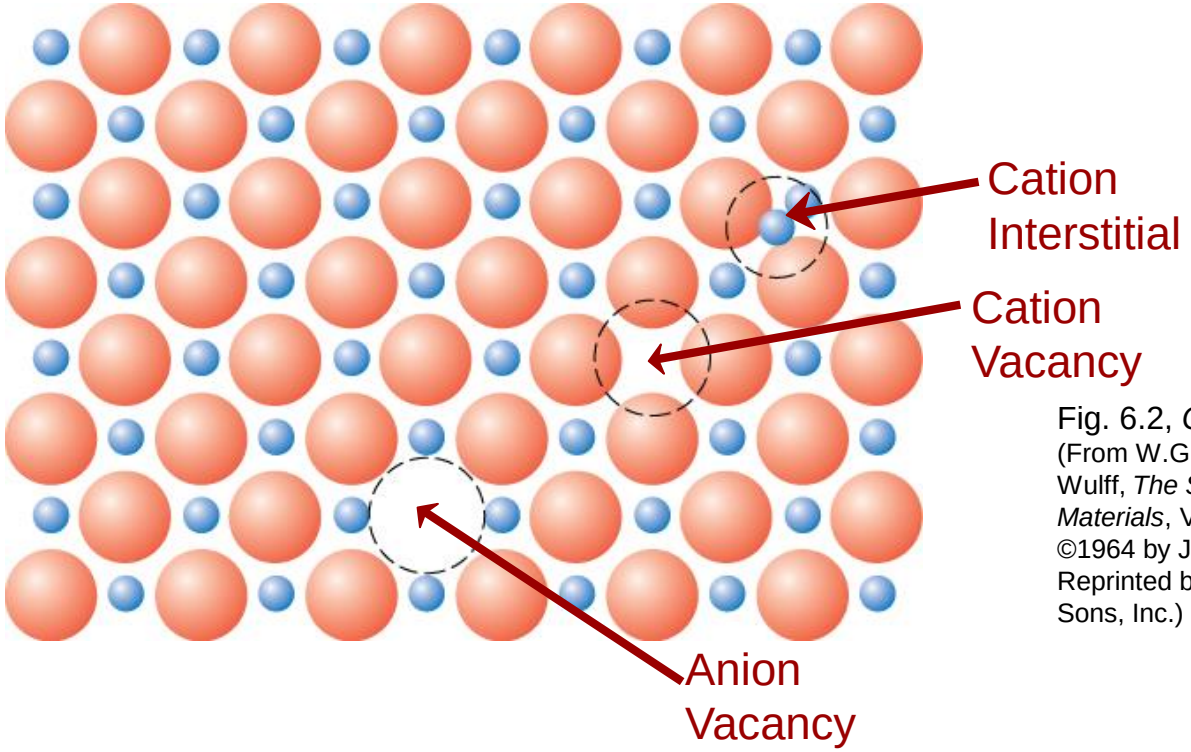
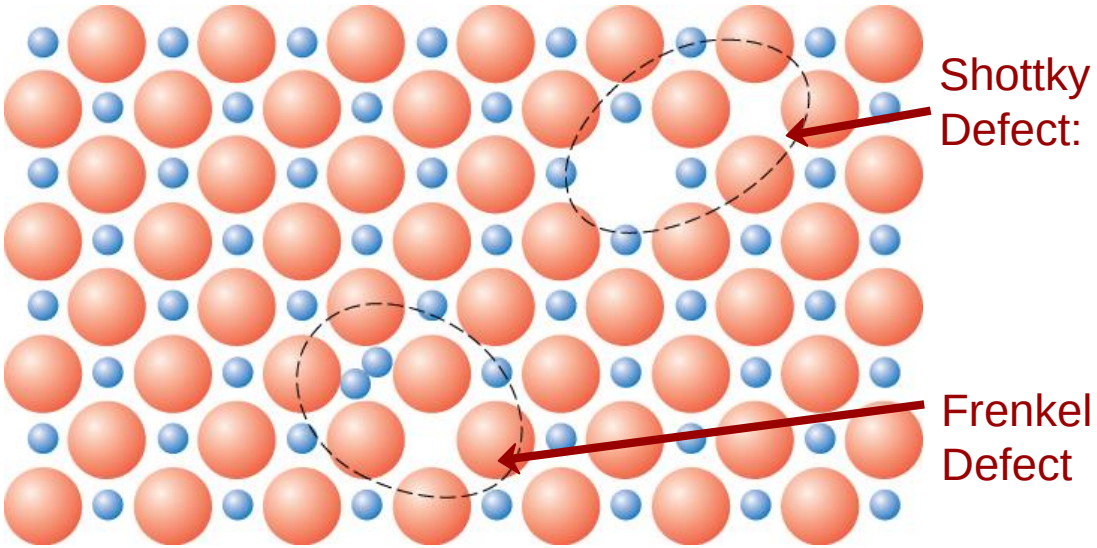


Fig. 6.2, *Callister & Rethwisch 9e.* (From W.G. Moffatt, G.W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. 1, *Structure*, p.78. Copyright ©1964 by John Wiley & Sons, New York. Reprinted by permission of John Wiley and Sons, Inc.)

- Frenkel Defect is a form of point defect that is created when an atom or cation leaves its original place in the lattice structure to create a vacancy while occupying another interstitial position within the solid crystal.

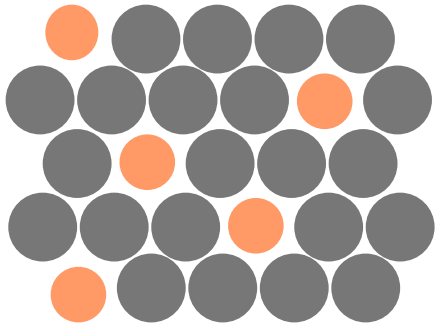
- Shottky Defect: is a point defect in which both cation and anion remain missing from the lattice site in equal numbers.
  -- a paired set of cation and anion vacancies.



Imperfections in Metals (i)

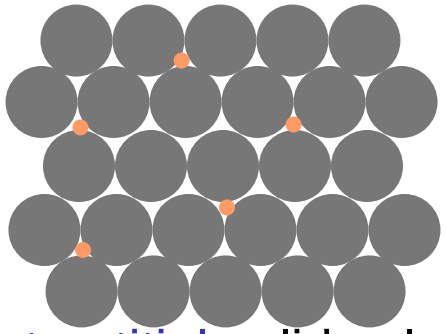
Two outcomes if impurity (B) added to host (A):

- Solid solution of B in A (i.e., random dist. of point defects)



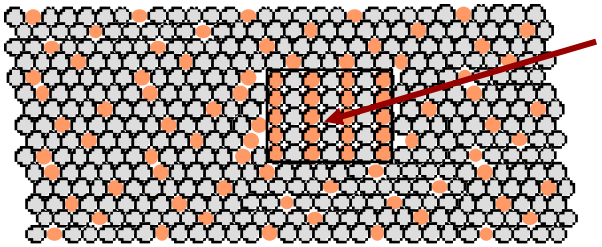
Substitutional solid soln.
(e.g., Cu in Ni)

OR



Interstitial solid soln.
(e.g., C in Fe)

- Solid solution of B in A plus particles of a new phase (usually for a larger amount of B)



Second phase particle
-- different composition
-- often different structure.

Imperfections in Metals (ii)

Conditions for substitutional solid solution (S.S.)

- W. Hume – Rothery rule
  - 1. Δr (atomic radius) < 15%
  - 2. Proximity in periodic table
    - i.e., similar electronegativities
  - 3. Same crystal structure for pure metals
  - 4. Valency
    - All else being equal, a metal will have a greater tendency to dissolve a metal of higher valency than one of lower valency

Imperfections in Metals (iii)

Application of Hume–Rothery rules – Solid Solutions

1. Would you predict more Al or Ag to dissolve in Zn?

2. More Zn or Al in Cu?

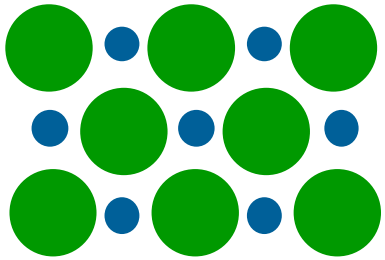
Element	*Atomic Radius (nm)*	*Crystal Structure*	*Electro-nega-tivity*	*Valence*
Cu	0.1278	FCC	1.9	+2
C	0.071			
H	0.046			
O	0.060			
Ag	0.1445	FCC	1.9	+1
Al	0.1431	FCC	1.5	+3
Co	0.1253	HCP	1.8	+2
Cr	0.1249	BCC	1.6	+3
Fe	0.1241	BCC	1.8	+2
Ni	0.1246	FCC	1.8	+2
Pd	0.1376	FCC	2.2	+2
Zn	0.1332	HCP	1.6	+2

Table on p. 177, *Callister & Rethwisch 9e.*

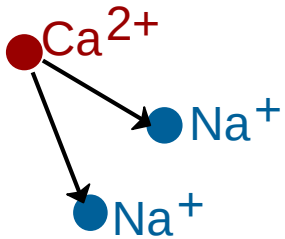
Imperfections in Ceramics

- Electroneutrality (charge balance) must be maintained when impurities are present
- Ex: NaCl 

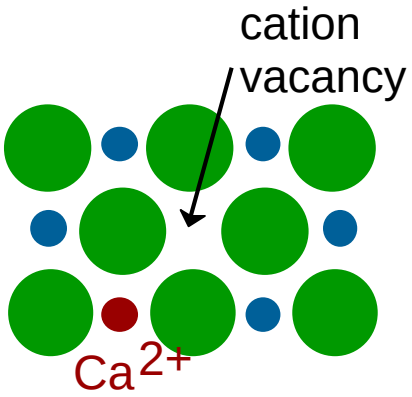
- Substitutional cation impurity



without impurity

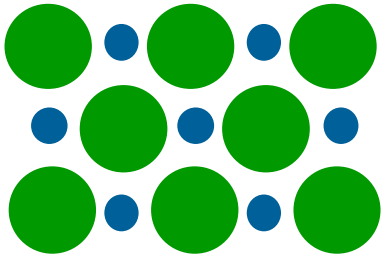


Ca^{2+} impurity

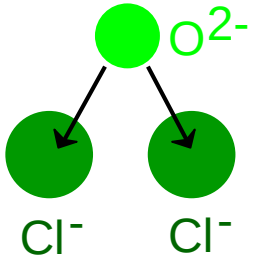


with impurity

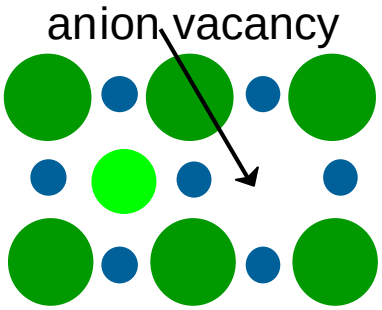
- Substitutional anion impurity



without impurity



O^{2-} impurity



with impurity

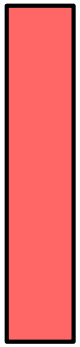
Line Defects

Dislocations:

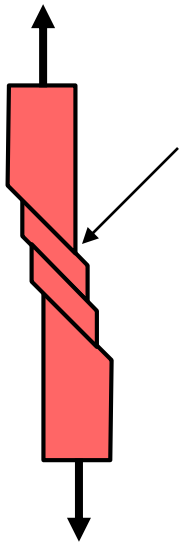
- are line defects,
- slip between crystal planes result when dislocations move,
- produce permanent (plastic) deformation.

Schematic of Zinc (HCP):

- before deformation



- after tensile elongation



Imperfections in Solids

Linear Defects (Dislocations)

- Are one-dimensional defects around which atoms are misaligned

- Edge dislocation:
  - extra half-plane of atoms inserted in a crystal structure
  - **b** perpendicular ($\perp$) to dislocation line

- Screw dislocation: a dislocation in the lattice structure of a crystal in which the atoms are arranged in a helical pattern that is normal to the direction of the stress. **b** parallel ($\parallel$) to dislocation line.

- Burger's vector, **b**: represents the magnitude and direction of the lattice distortion resulting from a dislocation in a crystal lattice.

Imperfections in Solids

Edge Dislocation

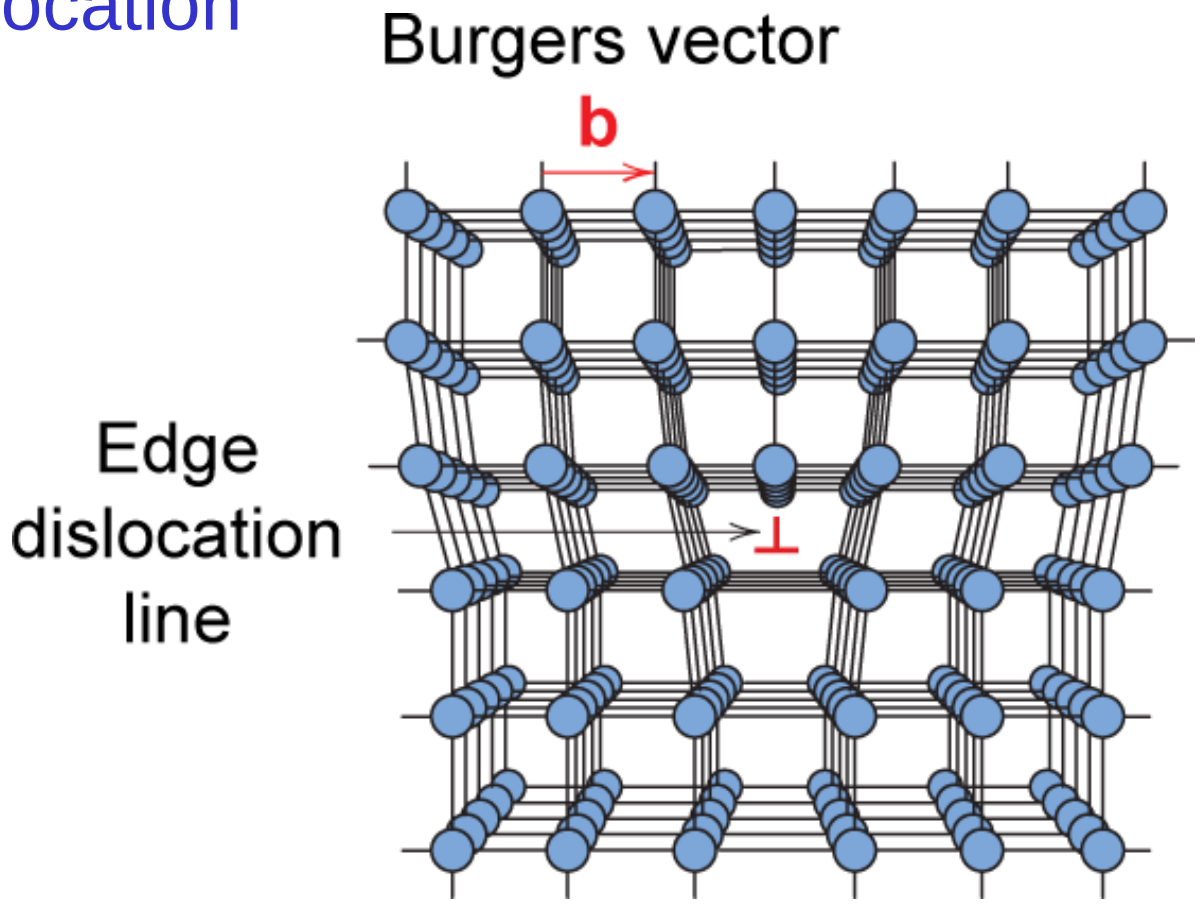
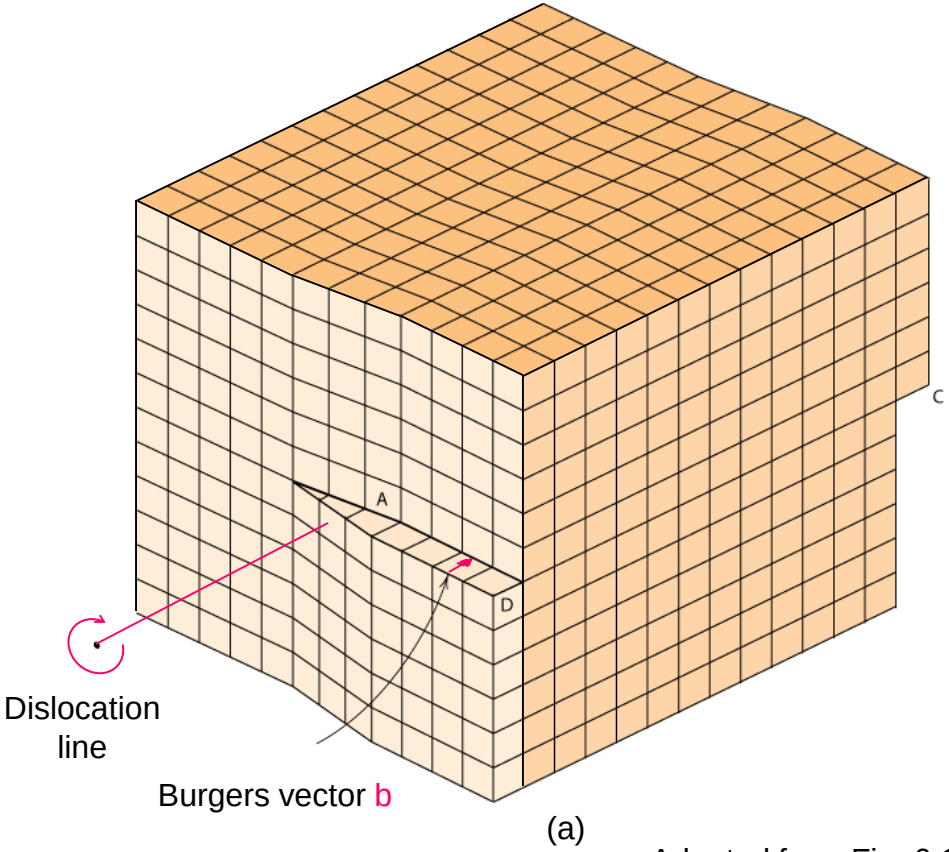


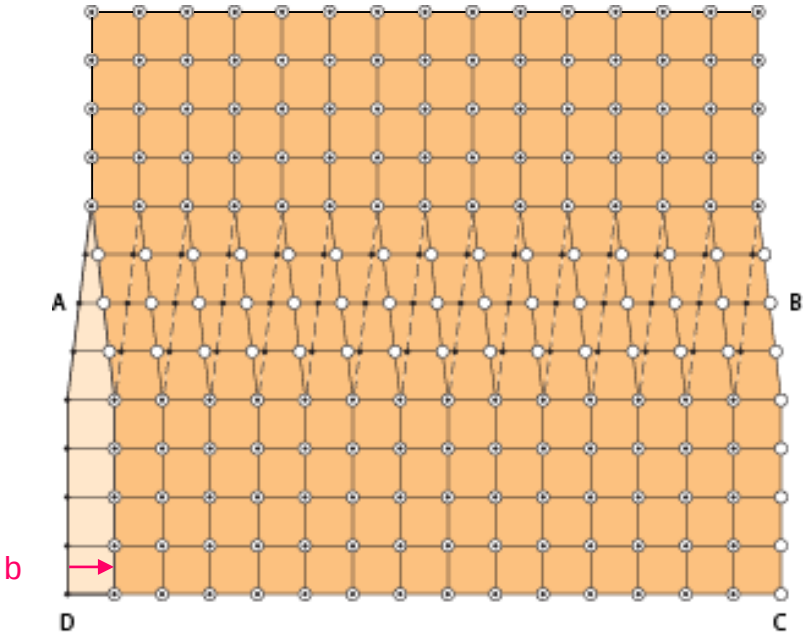
Fig. 6.9, *Callister & Rethwisch 9e.* (Adapted from A. G. Guy, Essentials of Materials Science, McGraw-Hill Book Company, New York, NY, 1976, p. 153.)

Imperfections in Solids

Screw Dislocation



(a)



(b)

Adapted from Fig. 6.10, *Callister & Rethwisch 9e.*
[Figure (b) from W. T. Read, Jr.,Dislocations in Crystals, McGraw-Hill Book Company, New York, NY, 1953.]

Imperfections in Solids

Dislocations are visible in electron micrographs

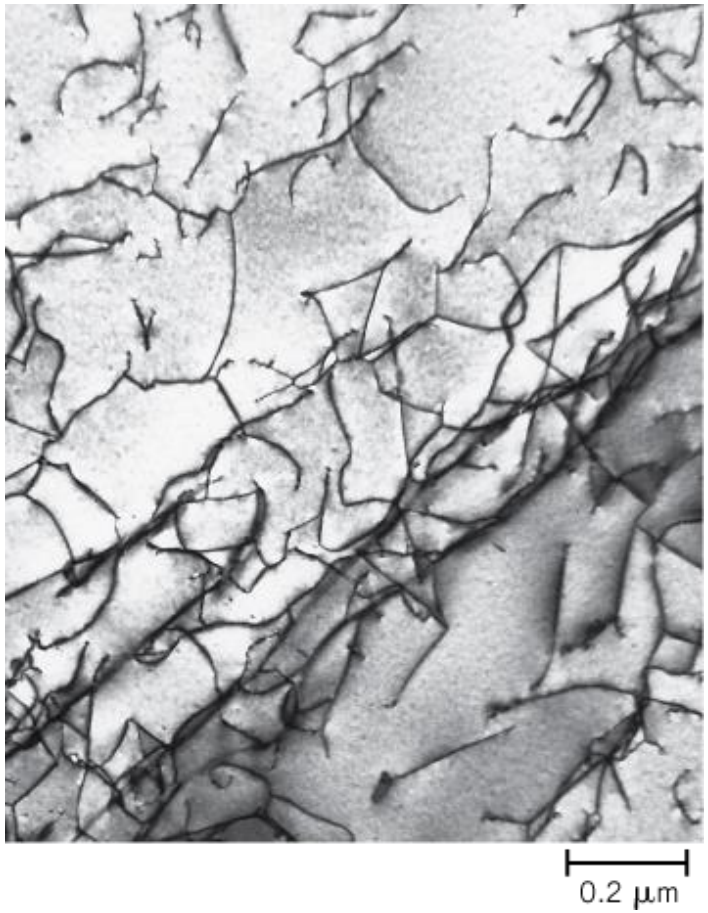
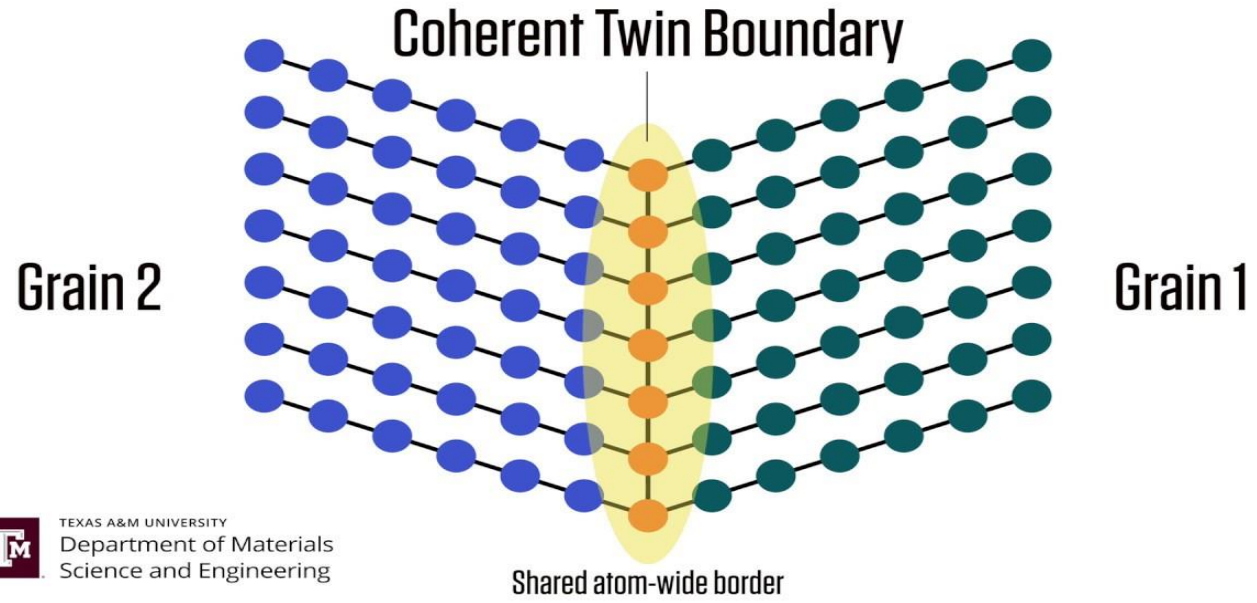


Fig. 6.12, *Callister & Rethwisch 9e.*
(Courtesy of M. R. Plichta, Michigan Technological University.)

Planar Defects in Solids

- *Twin boundary*: A grain boundary whose lattice structures are mirror images of each other in the plane of the boundary
  - Essentially a reflection of atom positions across the twin plane.



- Stacking faults: when there is an error in the stacking or packing sequence.
  - For FCC metals an error in ABCABC packing sequence
  - Ex: ABCABABC

Microscopic Examination

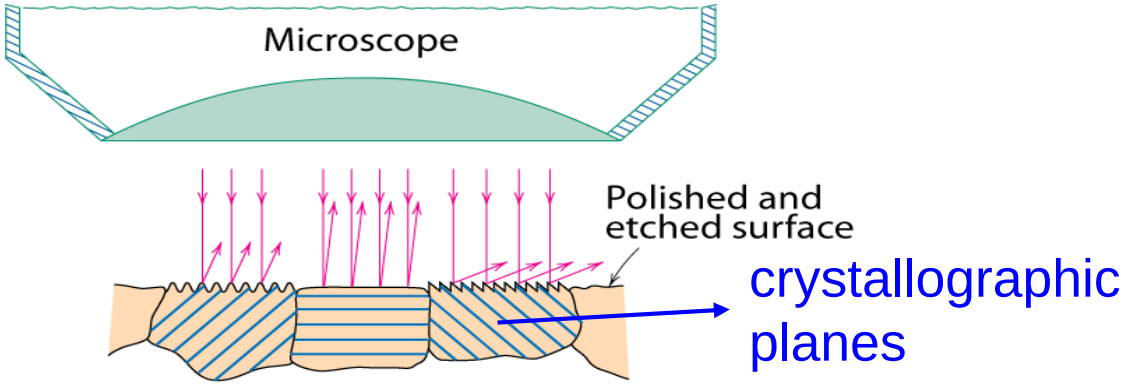
- Crystallites (grains) can be quite small (mm or less) – necessary to observe with a microscope.

Optical Microscopy

- Useful up to 2000X magnification.
- Polishing removes surface features (e.g., scratches)
- Etching changes reflectance, depending on crystal orientation.



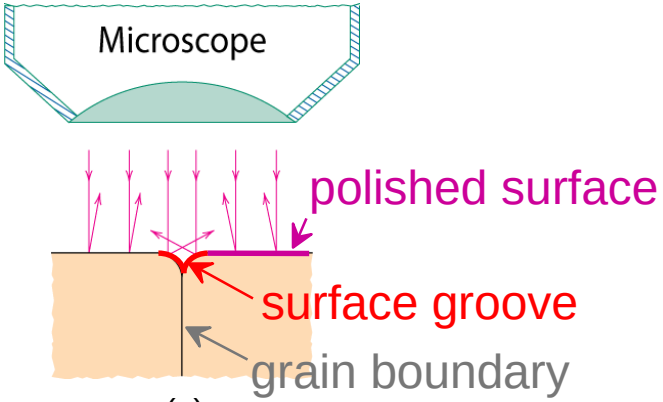
Micrograph of brass (a Cu-Zn alloy)



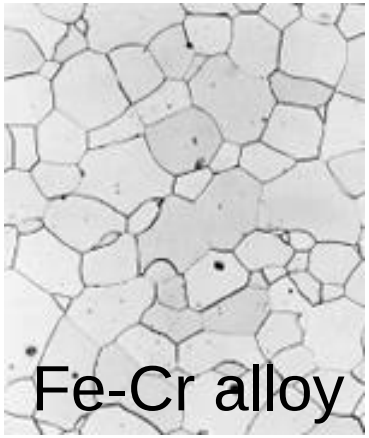
Optical Microscopy

Grain boundaries...

- are imperfections,
- are more susceptible to etching,
- may be revealed as dark lines,
- change in crystal orientation across boundary.



(a)



(b)

Fig. 6.20(a) & (b), *Callister & Rethwisch 9e.*
[Fig. 6.20(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD).]

Summary

- Point, Line, and Area defects exist in solids.
- The number and type of defects can be varied and controlled (e.g., temperature controls vacancy concentration).
- Defects affect material properties (e.g., grain boundaries control crystal slip).
- Defects may be desirable or undesirable (e.g., dislocations may be good or bad, depending on whether plastic deformation is desirable or not).

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: